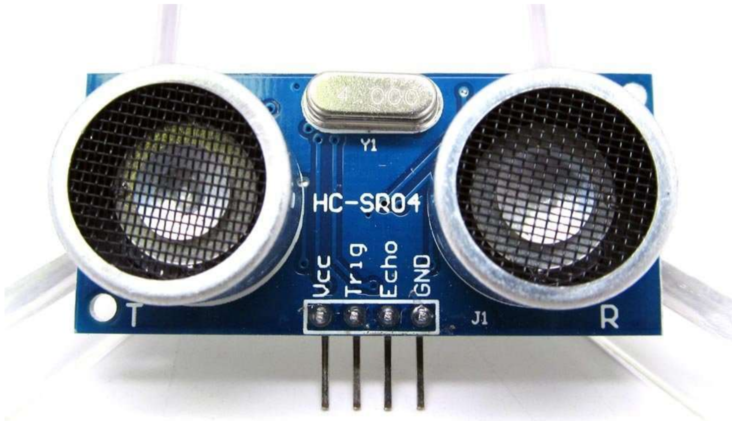
# Ultrasonic Distance Measurement Using HC-SR04 Sensor

The HC-SR04 ultrasonic sensor is widely used for non-contact distance measurement in robotic, industrial, and IoT applications. It operates by emitting ultrasonic waves and measuring the time it takes for the echo to return after reflecting off an object. With a measurement range of 2 cm to 400 cm and accuracy up to 3 mm, it provides a simple and effective solution for distance sensing.




## Working Principle

The HC-SR04 works based on the time-of-flight method. It sends out an ultrasonic pulse at 40 kHz frequency, waits for the echo to reflect from the object, and then calculates the distance based on the time difference between sending and receiving the pulse. The speed of sound in air (approximately 340 m/s) is used for this calculation.

The sensor requires a trigger signal and then measures the echo pulse width, which is directly proportional to the distance of the object.

## Interfacing and Wiring Details

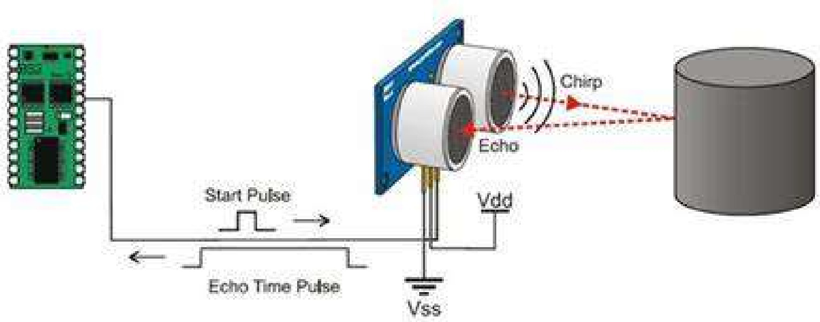
**Pin Configuration:**

- VCC: 5V DC power supply
- TRIG: Trigger input (initiates measurement)
- ECHO: Echo output (pulse width indicates distance)
- GND: Ground connection

**Connection :**

- The sensor should be powered via a 5V regulated supply.
- When wiring, always connect GND first before powering the module, to avoid malfunction.
- Avoid direct connection to high-voltage sources.
- The trigger input must be given a HIGH pulse of at least 10 μs to initiate a measurement.
- The module will automatically send out eight 40 kHz pulses and raise the ECHO pin HIGH.
- The duration of the HIGH pulse on ECHO corresponds to the round-trip time of the ultrasonic wave.

**Important Recommendation:**
Maintain a measurement cycle of at least 60 ms to prevent signal overlap, ensuring accurate readings.




**Timing Diagram and Pulse Sequence**

1. Trigger Signal: A HIGH pulse of minimum 10 μs is sent to TRIG.
2. Ultrasonic Burst: The sensor emits an 8-cycle burst at 40 kHz.
3. Echo Signal: If the signal reflects off an object, the ECHO pin goes HIGH for a time proportional to the distance.
4. Distance Calculation: Measure the pulse width of ECHO to compute distance.

**Measurement Conditions and Accuracy**

- Range: 2 cm to 400 cm
- Accuracy: 3 mm under optimal conditions
- Target Surface: Should be flat and smooth, with a minimum area of 0.5 square meters. Irregular or small surfaces can cause inaccurate results due to poor echo reflection.

**Typical Use Cases**

The HC-SR04 is commonly used in applications requiring real-time distance monitoring and object detection. Typical use cases include:

- Robotics: Obstacle avoidance, autonomous navigation
- IoT Projects: Smart parking systems, water level detection
- Industrial Automation: Proximity sensing on assembly lines
- Agriculture: Liquid level monitoring in tanks
- Consumer Electronics: Automatic door openers, proximity alarms